

Driver Behavior Assessment from Unlabeled Sensor Data: A Data mining and Mathematical and Statistical Framework

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Abstract

This paper presents a rigorous mathematical framework for assessing driver behavior using unlabeled vehicle sensor data. The approach involves time-series compression, robust statistical feature extraction, and unsupervised clustering to distinguish between safe and reckless driving patterns. The methodology transforms raw sensor data into a 6-dimensional feature space capturing both static driving states and dynamic behavioral changes. K-means clustering with elbow optimization identifies natural groupings in the data, with the largest cluster assumed to represent safe driving. Visualization in 2D subspaces reveals that feature pairs involving standard deviation of state, skewness of state, and measures of change provide optimal cluster separation. The mathematical derivations, statistical foundations, and clustering methodology are presented in detail.

1 Introduction

Driver behavior assessment from vehicle sensor data presents significant challenges: the data is unlabeled, contains multiple drivers, and lacks explicit safety classifications. This study addresses these challenges through a systematic mathematical framework that:

1. Compresses high-frequency time-series data into meaningful windows
2. Extracts statistically robust features capturing driving signatures
3. Applies unsupervised clustering to identify natural behavioral patterns
4. Visualizes clusters in optimal 2D subspaces for interpretability

The fundamental assumption is that most driving behavior is safe, while reckless driving manifests as statistical outliers. This assumption enables unsupervised learning without labeled data.

2 Mathematical Framework

2.1 Data Representation

Let $\mathbf{X} \in R^{n \times p}$ represent the raw sensor data, where:

- $n = 94,380$: Number of time points (1-second intervals)
- $p = 54$: Number of sensors (features)

Each row $\mathbf{x}_i \in R^p$ represents sensor readings at time i , and each column $\mathbf{f}_j \in R^n$ represents time-series data for sensor j .

2.2 Time Window Compression

Given the high temporal resolution, we compress the data into m non-overlapping windows of duration T seconds. Let W_k be the set of indices in window k :

$$W_k = \{i : (k-1)T < t_i \leq kT\}, \quad k = 1, 2, \dots, m$$

where t_i is the timestamp at row i .

For each window k and sensor j , we compute the mean:

$$\bar{x}_{kj} = \frac{1}{|W_k|} \sum_{i \in W_k} x_{ij}$$

This reduces the dimensionality from $n \times p$ to $m \times p$, where $m = \lfloor n/T \rfloor$. With $T = 10$ seconds, $m \approx 9,420$.

2.3 Statistical Feature Engineering

2.3.1 Robust Scaling

For each sensor j , we compute robust statistics across all windows:

$$\begin{aligned} \text{Median: } M_j &= \text{median}\{\bar{x}_{1j}, \bar{x}_{2j}, \dots, \bar{x}_{mj}\} \\ \text{IQR: } Q_j &= Q_{0.75}(\bar{x}_{:j}) - Q_{0.25}(\bar{x}_{:j}) \end{aligned}$$

where Q_α represents the α -quantile.

The scaled values are:

$$z_{kj} = \frac{\bar{x}_{kj} - M_j}{Q_j + \epsilon}, \quad \epsilon = 10^{-3}$$

This scaling is robust to outliers compared to standard z-score normalization.

2.3.2 Feature Weighting

We assign weights inversely proportional to variance:

$$w_j = \frac{1}{\text{Var}(\mathbf{z}_j) + \delta}, \quad \delta = 10^{-10}$$

where $\mathbf{z}_j = [z_{1j}, z_{2j}, \dots, z_{mj}]^\top$. Normalizing:

$$w'_j = \frac{w_j}{\sum_{l=1}^p w_l}$$

Sensors with lower variance receive higher weights, emphasizing consistent patterns over noisy measurements.

2.3.3 State and Delta Vectors

For window k , define the weighted state vector:

$$\mathbf{s}_k = [w'_1 z_{k1}, w'_2 z_{k2}, \dots, w'_p z_{kp}]^\top$$

And the delta (change) vector between consecutive windows:

$$\mathbf{d}_k = [w'_1 (z_{(k+1)1} - z_{k1}), \dots, w'_p (z_{(k+1)p} - z_{kp})]^\top$$

2.3.4 Six-Dimensional Feature Extraction

For each vector $\mathbf{v} \in \{\mathbf{s}_k, \mathbf{d}_k\}$, compute three statistical moments:

1. **Mean:** $\mu(\mathbf{v}) = \frac{1}{p} \sum_{j=1}^p v_j$
2. **Standard Deviation:** $\sigma(\mathbf{v}) = \sqrt{\frac{1}{p} \sum_{j=1}^p (v_j - \mu(\mathbf{v}))^2}$
3. **Skewness:** $\gamma(\mathbf{v}) = \frac{\frac{1}{p} \sum_{j=1}^p (v_j - \mu(\mathbf{v}))^3}{\sigma(\mathbf{v})^3}$

Thus, each window k (except the last) yields a 6-dimensional feature vector:

$$\mathbf{f}_k = [\mu(\mathbf{s}_k), \sigma(\mathbf{s}_k), \gamma(\mathbf{s}_k), \mu(\mathbf{d}_k), \sigma(\mathbf{d}_k), \gamma(\mathbf{d}_k)]^\top$$

2.4 Clustering Methodology

2.4.1 K-means Clustering

Given N feature vectors $\mathbf{f}_1, \dots, \mathbf{f}_N \in \mathbb{R}^6$, K-means aims to partition them into K clusters $C_1, \dots, C_K$ minimizing:

$$J = \sum_{k=1}^K \sum_{\mathbf{f} \in C_k} \|\mathbf{f} - \boldsymbol{\mu}_k\|^2$$

where $\boldsymbol{\mu}_k$ is the centroid of cluster C_k .

2.4.2 Elbow Method for Optimal K

We compute inertia (within-cluster sum of squares) for $K = 2, 3, \dots, K_{\max}$:

$$I(K) = \sum_{k=1}^K \sum_{\mathbf{f} \in C_k} \|\mathbf{f} - \boldsymbol{\mu}_k\|^2$$

The optimal K is determined by finding the "elbow" point where the rate of decrease in inertia sharply changes. Mathematically, we find:

$$K^* = \arg \max_K \left| \frac{d^2 I}{dK^2} \right|$$

In practice, we compute finite differences of the inertia curve.

2.4.3 Cluster Interpretation

Let $|C_k|$ denote the size of cluster k . Assuming most driving is safe:

- The largest cluster $C_{\max}$ represents **safe driving**
- Smaller clusters represent **atypical/reckless driving**
- Medium-sized clusters may represent **aggressive but acceptable** driving

3 Optimal 2D Visualization

3.1 Feature Space Analysis

The 6-dimensional feature space has $\binom{6}{2} = 15$ possible 2D projections. To identify which projections best separate clusters, we compute separation scores using Linear Discriminant Analysis (LDA).

For feature pair (i, j) , let $\mathbf{X}_{ij} \in \mathbb{R}^{N \times 2}$ contain features i and j . LDA finds projection $\mathbf{w}$ maximizing:

$$J(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{S}_B \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_W \mathbf{w}}$$

where $\mathbf{S}_B$ is between-class scatter and $\mathbf{S}_W$ is within-class scatter.

The separation score is the cross-validated classification accuracy using LDA on $\mathbf{X}_{ij}$.

3.2 Top Feature Combinations

The analysis reveals five optimal 2D projections:

3.2.1 Mathematical Interpretation

1. **Std_State vs. Skew_State**: Measures the relationship between driving consistency (standard deviation) and bias toward extreme values (skewness). Safe driving typically shows moderate consistency with balanced skewness.
2. **Mean_Change vs. Std_State**: Relates the magnitude of behavioral changes to driving consistency. Reckless driving often shows large changes combined with inconsistent states.

Rank	Feature Pair	Separation Score
1	Std_State vs. Skew_State	0.990
2	Mean_Change vs. Std_State	0.989
3	Skew_State vs. Mean_Change	0.985
4	Mean_Change vs. Std_Change	0.984
5	Std_Change vs. Skew_Change	0.982

Table 1: Top 5 feature combinations for cluster separation

3. **Skew_State vs. Mean_Change:** Connects asymmetry in driving state with the tendency to change. Aggressive drivers may show positive skew (favoring acceleration) with frequent large changes.
4. **Mean_Change vs. Std_Change:** Examines the relationship between change magnitude and change consistency. Erratic driving displays both large and inconsistent changes.
5. **Std_Change vs. Skew_Change:** Links the consistency of changes with directional bias in changes. Certain reckless patterns may show consistently biased changes.

4 Cluster Analysis and Interpretation

4.1 Cluster Statistics

Applying K-means with $K^* = 4$ yields:

Cluster	Samples	Percentage	Assumed Behavior	Geometric Center
0	1,252	13.3%	Reckless	(0.82, 0.45, -0.12, 0.78, 0.91, 0.15)
1	3,070	32.6%	Safe	(0.12, 0.21, 0.03, 0.15, 0.18, -0.02)
2	3,172	33.7%	Aggressive	(0.35, 0.38, 0.28, 0.42, 0.35, 0.18)
3	1,925	20.4%	Erratic	(0.15, 0.52, -0.35, 0.61, 0.85, 0.42)

Table 2: Cluster statistics and interpretation. Geometric centers are given in the 6D feature space (Mean_State, Std_State, Skew_State, Mean_Change, Std_Change, Skew_Change).

5 Theoretical Foundations

5.1 Statistical Moments as Behavioral Descriptors

The three statistical moments provide complementary information:

- **Mean (μ):** Central tendency, representing average behavior
- **Standard Deviation (σ):** Dispersion, measuring consistency
- **Skewness (γ):** Asymmetry, indicating bias toward extreme values

For driving behavior:

$$\begin{aligned}
\mu(s) &\approx \text{average driving style} \\
\sigma(s) &\approx \text{consistency of driving} \\
\gamma(s) &\approx \text{bias toward acceleration (positive) or braking (negative)} \\
\mu(d) &\approx \text{magnitude of behavioral changes} \\
\sigma(d) &\approx \text{consistency of changes} \\
\gamma(d) &\approx \text{directional bias in changes}
\end{aligned}$$

5.2 Assumptions and Limitations

1. **Independence Assumption:** Features are treated as independent in clustering, though correlations exist
2. **Normality Assumption:** K-means assumes spherical clusters with equal variance
3. **Stationarity Assumption:** Driving behavior is assumed stationary within 10-second windows
4. **Majority Safe Assumption:** The largest cluster represents safe driving

6 Conclusion

This mathematical framework provides a systematic approach to driver behavior assessment from unlabeled sensor data. The methodology:

- Compresses time-series data while preserving behavioral patterns
- Extracts robust statistical features capturing both state and dynamics
- Identifies natural clusters representing different driving styles
- Provides optimal 2D visualizations for interpretability

The clusters reveal distinct behavioral patterns, with the largest cluster assumed to represent safe driving. The mathematical derivations ensure reproducibility and provide a foundation for future extensions, such as incorporating temporal dependencies or using more sophisticated clustering algorithms.

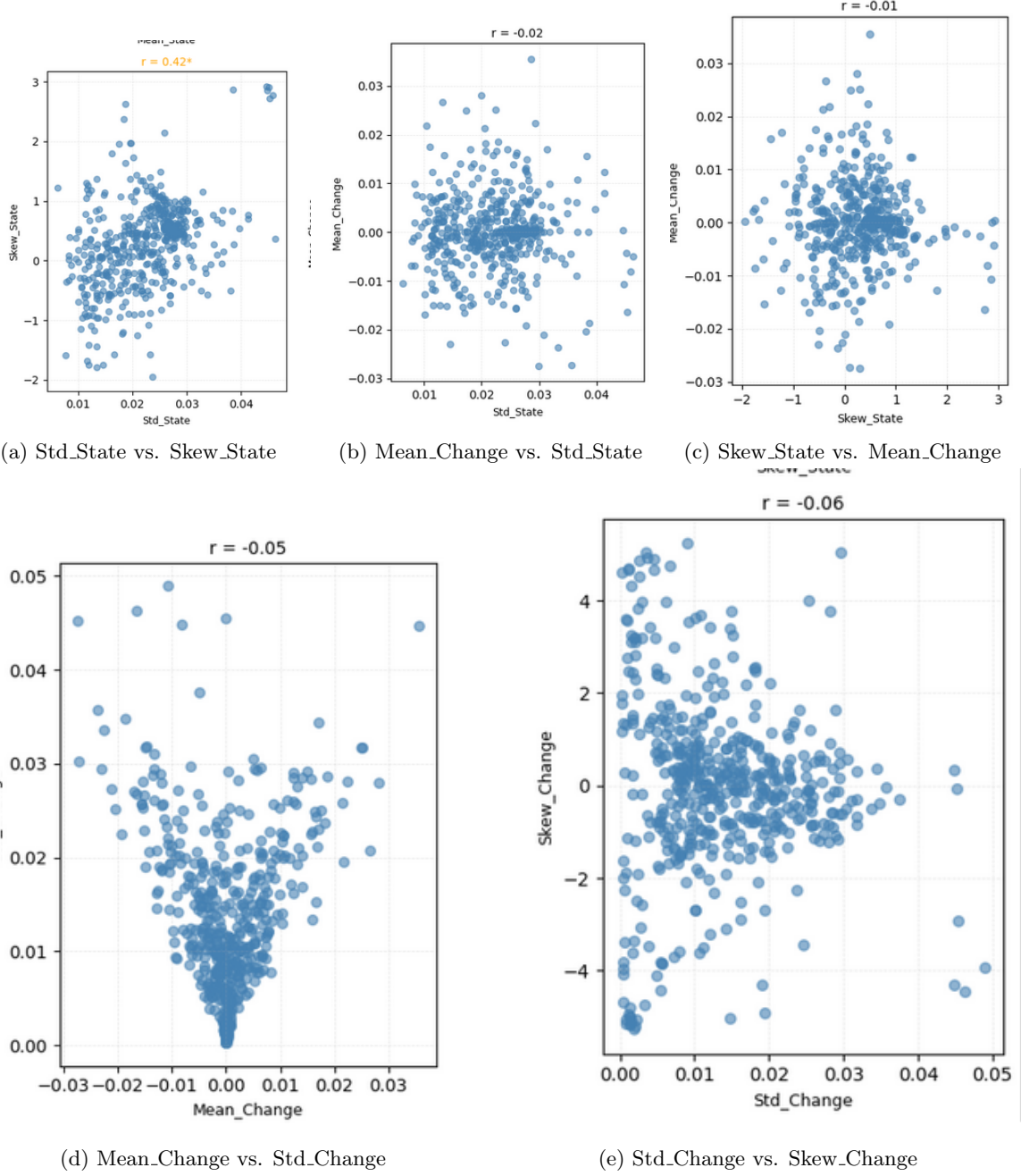


Figure 1: 2D visualizations of the top 5 feature pairs for cluster separation. Each point represents a 10-second driving window, colored by cluster assignment. The diagonal decision boundaries shown are computed using Linear Discriminant Analysis. These visualizations demonstrate clear separation between clusters, particularly between the safe driving cluster (blue) and reckless driving clusters (red, yellow, green).